VO VAN TRINH

Senior Backend Engineer / Technical Lead

Phone: (+84) 393 078 230 | Email: vovantrinh1992@gmail.com | Location: Ho Chi Minh City, Vietnam

PROFESSIONAL SUMMARY

Senior Software Engineer and Technical Lead with 10+ years of experience specializing in high-concurrency backend architecture, distributed systems, ERP customization, and real-time enterprise platforms. Experienced in applying SOLID principles, Clean Architecture, and proven design patterns to build maintainable, testable, and extensible systems. Proven track record of leading teams of 4–8 engineers, migrating legacy systems to microservices, and delivering OMS, ERP, POS, CRM, and product traceability platforms.

TECHNICAL SKILLS

- Languages & Frameworks: Python (Frappe Framework, ERPNext), TypeScript/JavaScript (Node.js, NestJS, Next.js, Electron.js), PHP (Laravel, Lumen, CodeIgniter).
- Databases & Storage: PostgreSQL, MySQL, MariaDB, MongoDB, Redis, Elasticsearch, CEPH (S3-compatible distributed storage).
- Architecture & System Design: SOLID Principles, Clean Architecture, Microservices, Event-Driven Architecture, Domain-Driven Design, RESTful APIs, WebSocket, Distributed Systems.
- Design Patterns: Repository, Factory, Strategy, Adapter, Facade, Observer, Dependency Injection, Unit of Work, CQRS, and MVC.
- DevOps & Infrastructure: Docker, Kubernetes, CI/CD, Git, Linux/Unix, CDN, and Cloud Storage.
- Leadership: Technical Roadmaps, Architecture Reviews, Code Reviews, Mentoring, Task Planning, and Technical Direction.

WORK EXPERIENCE

S3Work Technology Co., Ltd. | Senior Software Engineer cum Technical Lead (Feb 2020 – Present)

- Led and mentored a team of 6–8 engineers, providing technical direction, task planning, architecture reviews, and code reviews.
- Applied SOLID principles, Clean Architecture, dependency injection, and appropriate design patterns to improve modularity, testability, and long-term maintainability.
- Co-architected migration of legacy monolithic systems to scalable NestJS-based microservices deployed with Docker and Kubernetes.
- Led delivery of xCRM, ERPNext custom applications, license management, and multi-channel POS/OMS platforms with real-time synchronization.
- Resolved backend performance bottlenecks involving database queries, third-party integrations, caching, and real-time communication.

KDATA Co., Ltd. | Senior Software Engineer (Jul 2017 – Jan 2020)

- Designed and deployed an S3-compatible distributed object storage system using CEPH.
- Built and optimized an internal Content Delivery Network for static asset delivery and edge caching.
- Extended WHMCS modules to automate customer provisioning, server/domain lifecycle, billing, and payment workflows.
- Conducted technical interviews, onboarding, and training for junior backend engineers.

PTIT Trading & Services / Freelance / Global Media Group | Web Developer (Aug 2015 – Jun 2017)

- Built custom CMS platforms and commercial web applications using PHP, MySQL, and WordPress.
- Integrated payment gateways and optimized performance for high-traffic client portals.

KEY PROJECTS

ERPNext Custom Modules & Python Connector Applications (Jun 2026 – Present) | Technical Lead & Backend Developer

Technologies: Python, Frappe Framework, ERPNext, MariaDB, REST API, Redis, Docker

- Designed custom ERPNext modules and standalone Python connectors for integration with external enterprise platforms.
- Implemented custom DocTypes, server-side business logic, and REST/webhook integrations for inventory and sales order synchronization.

Product Origin Traceability & Supply Chain Tracking System (May 2026 – Jun 2026) | Technical Lead & Backend Developer

Technologies: Python, NestJS, PostgreSQL, QR Code Processing, Docker

- Designed a product-origin traceability platform supporting supply-chain verification and anti-counterfeiting workflows.
- Built batch QR-code generation and validation APIs with auditable records across distribution checkpoints.

Ambassador Affiliate & Commission Platform (Jan 2025 – Present) | Technical Lead & Backend Architect (Team Size: 5)

Technologies: TypeScript, NestJS, PHP/Laravel, MongoDB, MySQL, Socket.io, Docker, Kubernetes

- Built an affiliate platform covering multi-level commission calculation, inventory synchronization, and order management.
- Implemented event-driven processing to control concurrency in stock allocation and balance reconciliation.

Omni-Channel Order Management System (OMS) (Sep 2025 – Jan 2026) **| Technical Lead & Backend Developer (Team Size: 4)**

Technologies: NestJS, TypeScript, Next.js, PostgreSQL, WebSocket, Docker

- Integrated ShopeeFood, GrabFood, and Haravan order streams with automated electronic invoice issuance.
- Designed resilient webhook processing with retries and idempotency controls.

S3FnB & Tra La Windows Desktop POS System (Feb 2020 – Jul 2024) **| Technical Lead (Team Size: 4–8)**

Technologies: Electron.js, Node.js, NestJS, Laravel, MySQL, Socket.io

- Developed an offline-first POS application with invoice printing, peripheral support, and real-time cloud synchronization.

EDUCATION

University of Information Technology (UIT) – Vietnam National University, HCMC

Bachelor of Engineering in Computer Networks & Communications (Sep 2010 – Oct 2015)